

8086 Microprocessor in Maximum Mode

Q. Explain the operation of the 8086 microprocessor in Maximum Mode. Discuss the role of the Bus Controller (8288) and how a multiprocessor environment is managed.

> **Definition to Remember**

1. Maximum Mode Configuration

In this mode, pins that previously generated control signals (like `ALE`, `WR`) take on new functions related to bus arbitration and multiprocessor coordination.

2. Reassigned Maximum Mode Signals

- **S2, S1, S0 (Status Lines):** The 8086 outputs its current cycle status on these 3 pins. The 8288 reads them to generate command signals.

3. Role of the 8288 Bus Controller

The 8288 decodes the `S2-S0` signals to generate system controls:

4. Managing the Multiprocessor Environment

- **Bus Arbitration:** To prevent data corruption, only one processor can use the bus at a time. A coprocessor asserts **RQ**. The 8086 finishes its cycle, floats its buses (High-Z), and asserts **GT** to hand over control.

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

> `Max
bus re